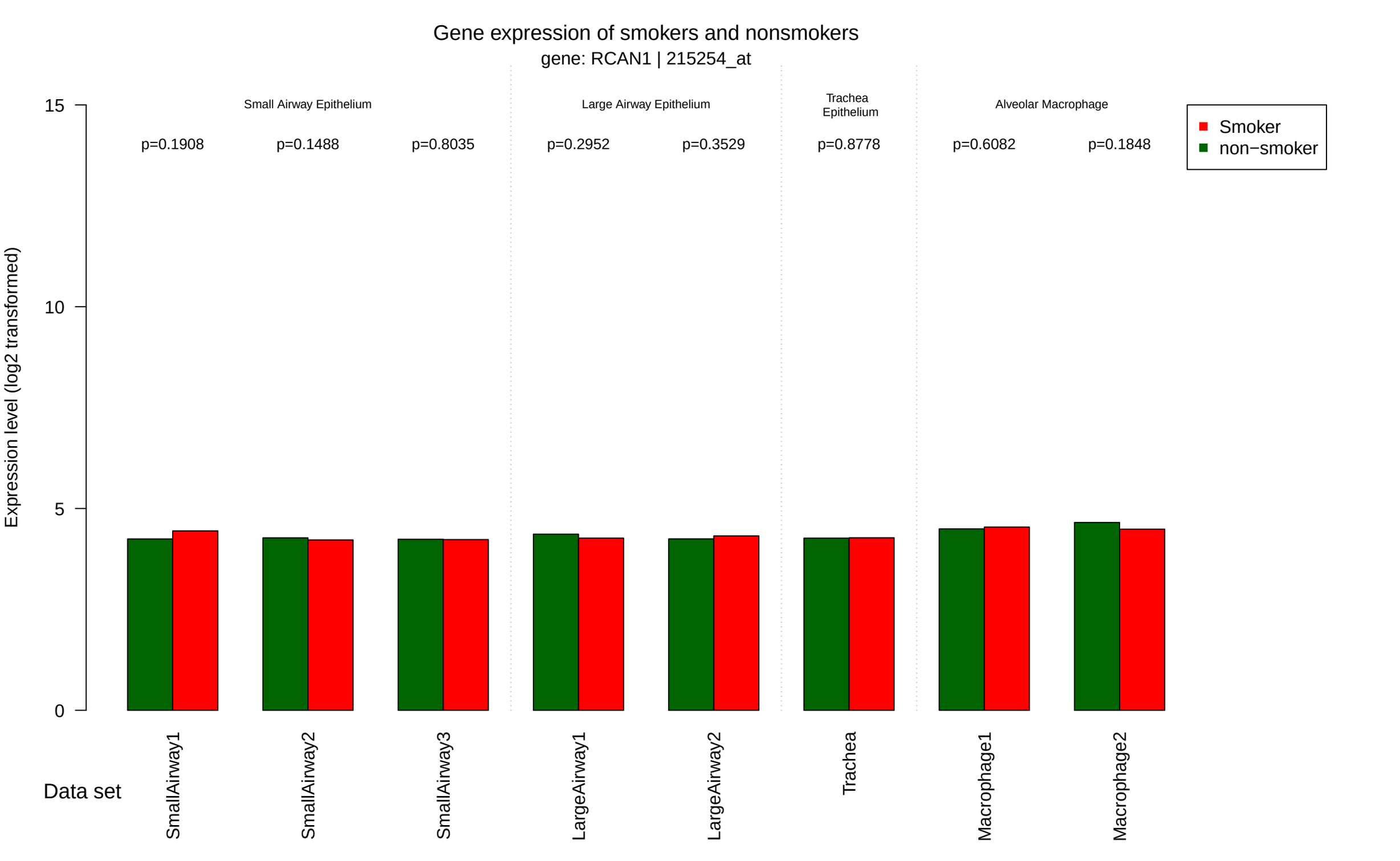
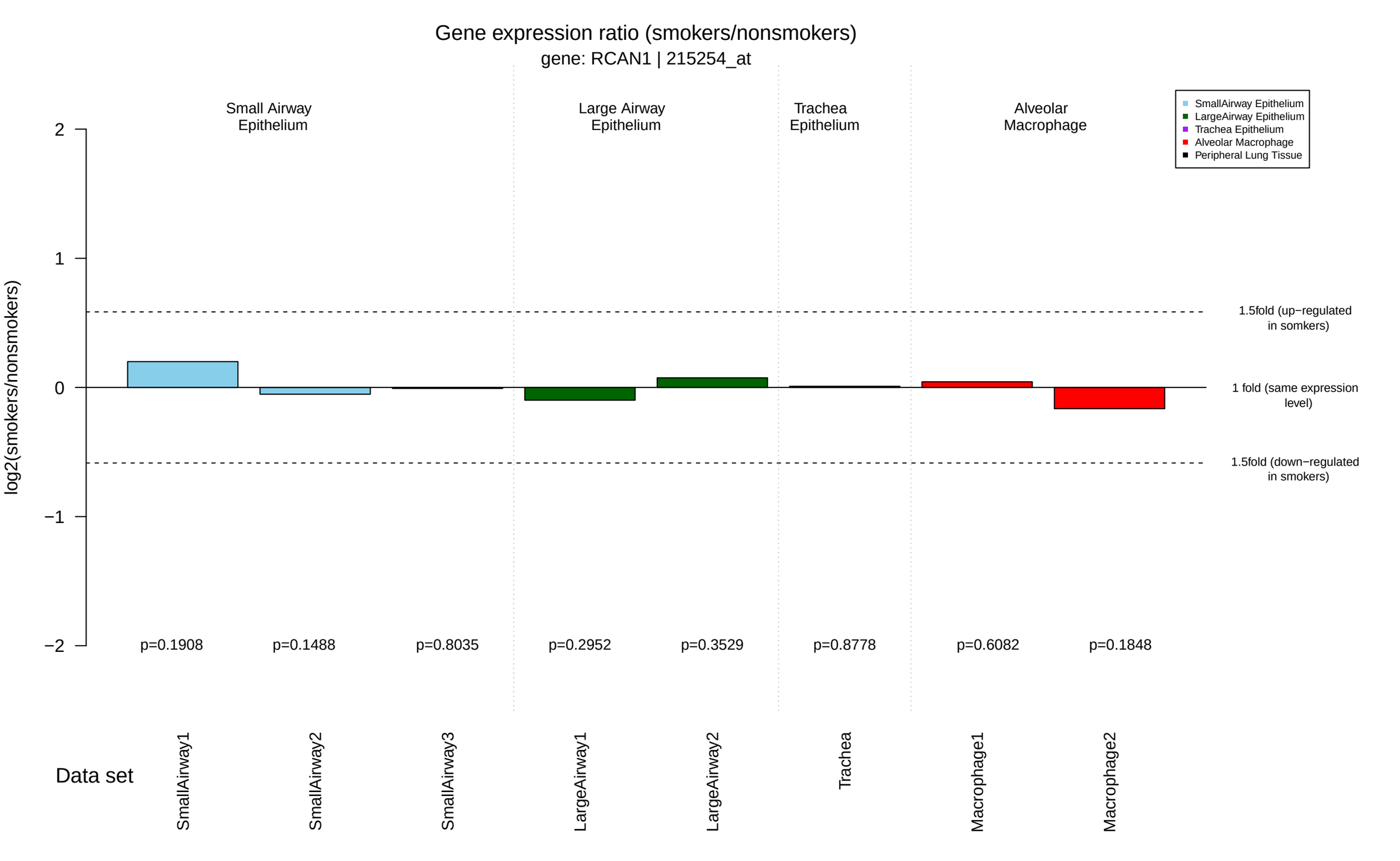
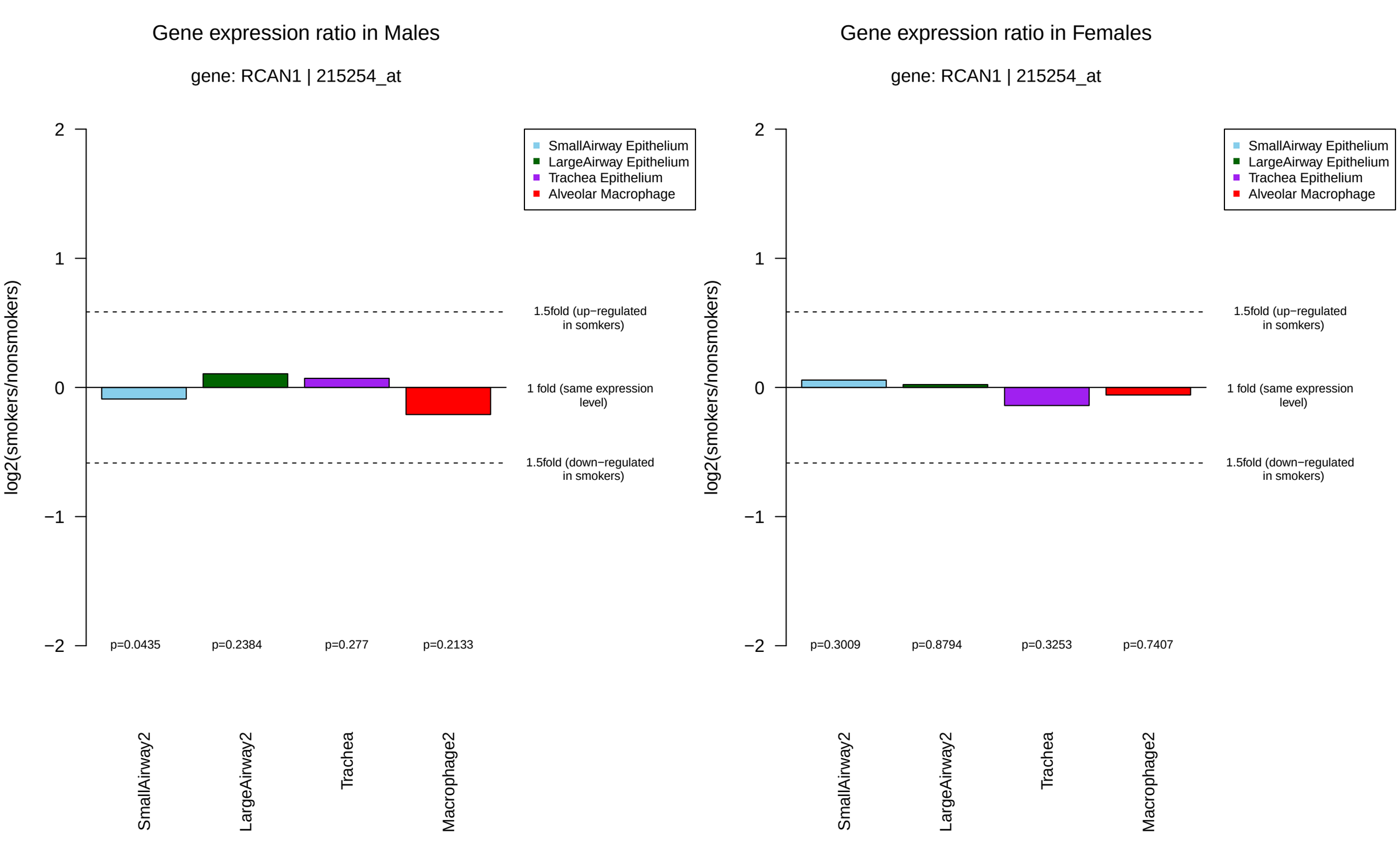
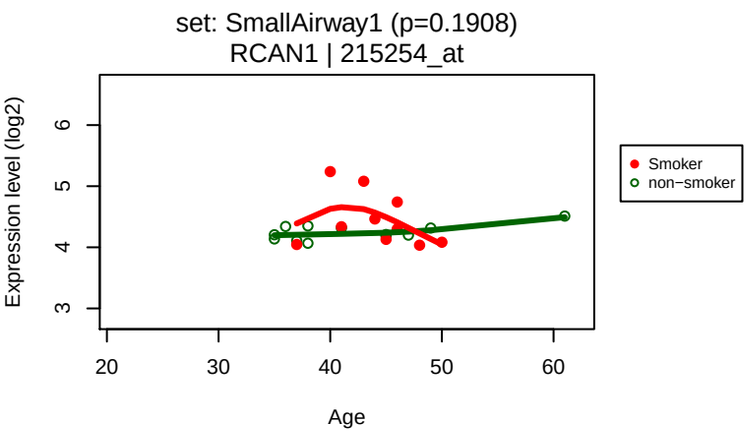
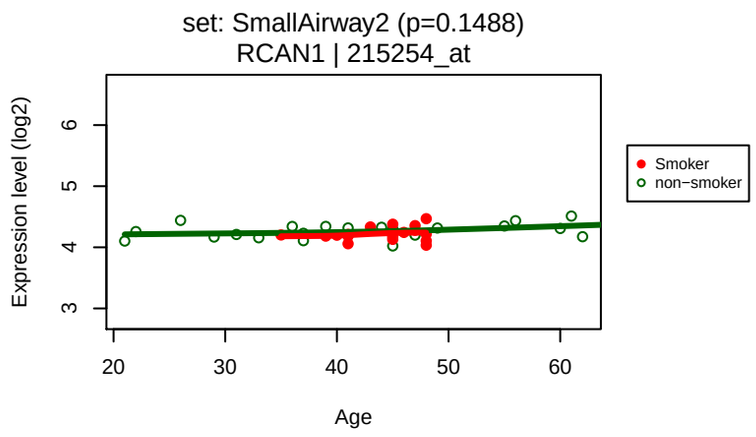
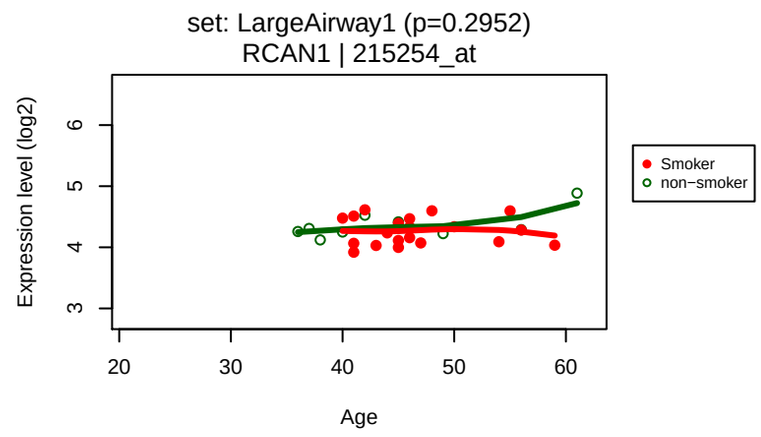

Gene expression of smokers and nonsmokers
gene: RCAN1 | 215254_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1908
p=0.1488
p=0.8035
p=0.2952
p=0.3529
p=0.8778
p=0.6082
p=0.1848
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

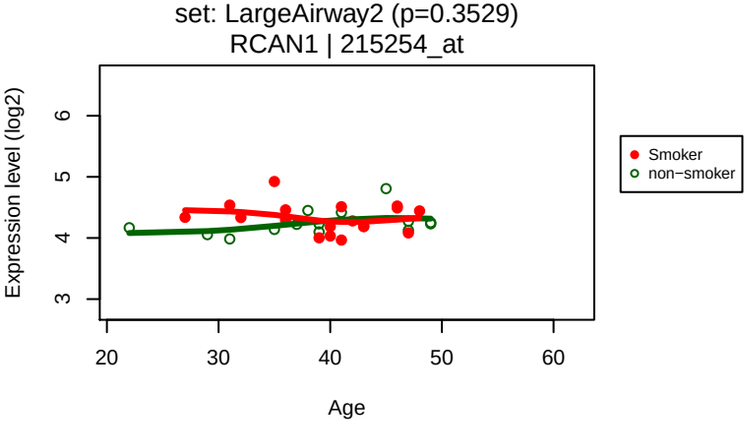

Gene expression ratio (smokers/nonsmokers)
gene: RCAN1 | 215254_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1908
p=0.1488
p=0.8035
p=0.2952
p=0.3529
p=0.8778
p=0.6082
p=0.1848
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

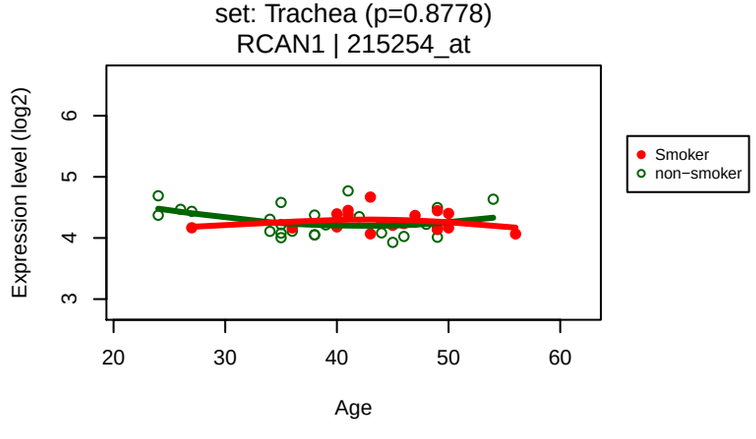

Gene expression ratio in Males
gene: RCAN1 | 215254_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0435
p=0.2384
p=0.277
p=0.2133
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RCAN1 | 215254_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3009
p=0.8794
p=0.3253
p=0.7407
SmallAirway2
LargeAirway2
Trachea
Macrophage2

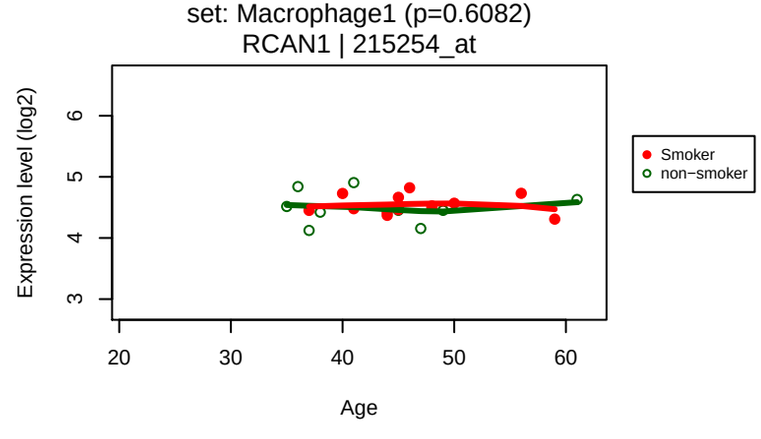

set: SmallAirway1 (p=0.1908)
RCAN1 | 215254_at
Expression level (log2)
Age
Smoker
non−smoker

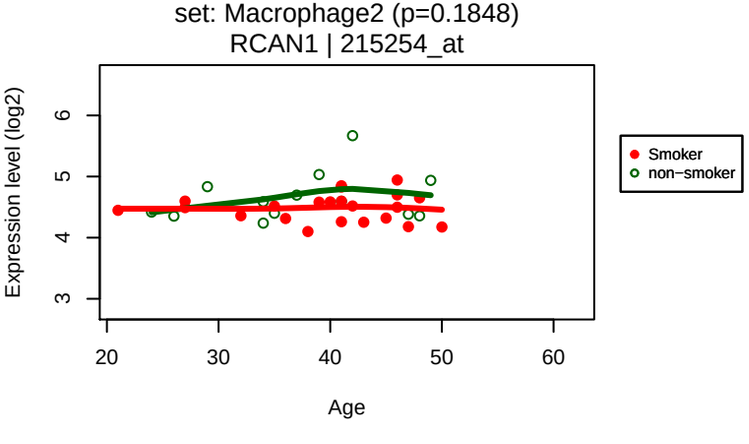

set: SmallAirway2 (p=0.1488)
RCAN1 | 215254_at
Expression level (log2)
Age
Smoker
non−smoker

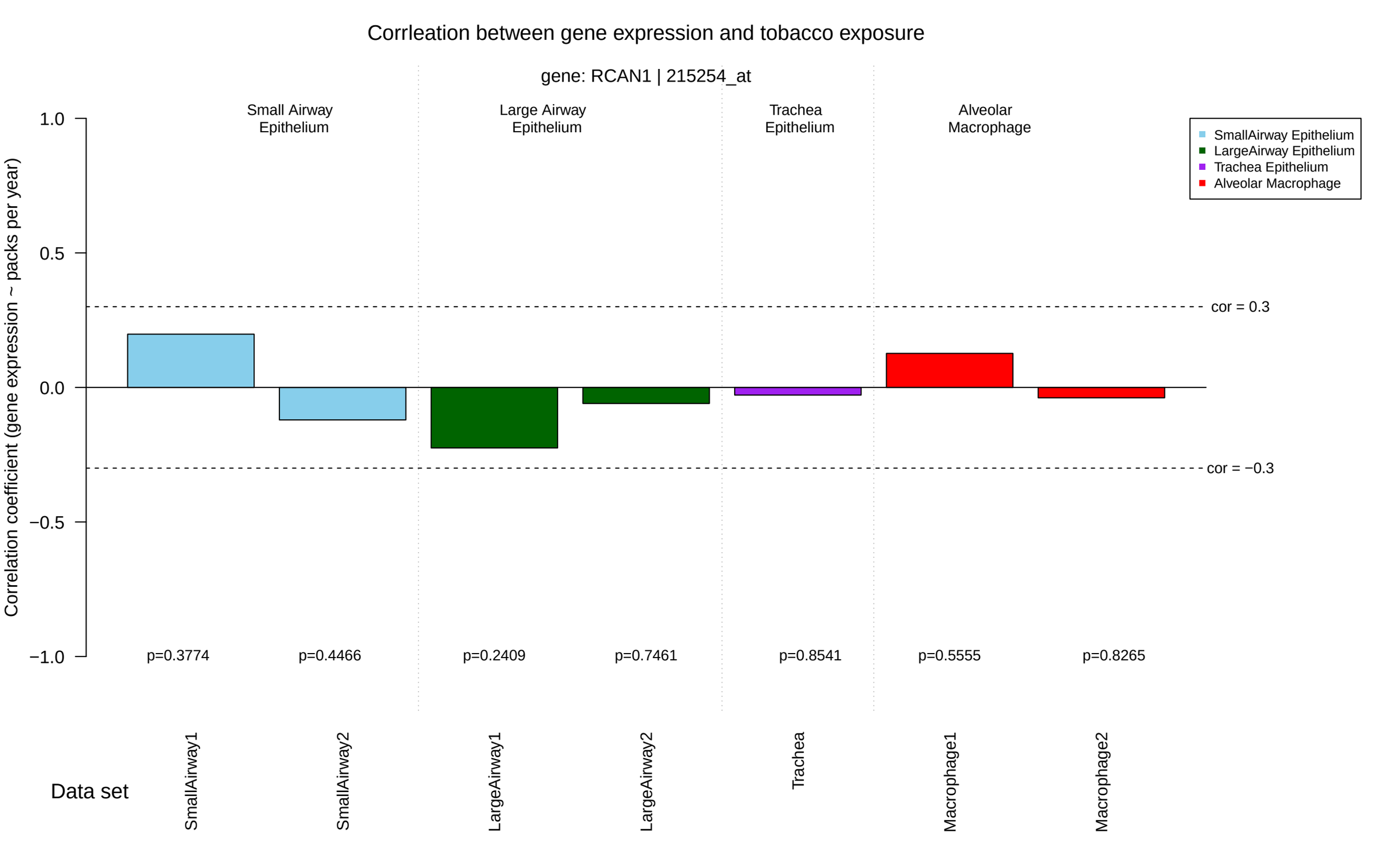

set: LargeAirway1 (p=0.2952)
RCAN1 | 215254_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3529)
RCAN1 | 215254_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8778)
RCAN1 | 215254_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6082)
RCAN1 | 215254_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1848)
RCAN1 | 215254_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RCAN1 | 215254_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3774
p=0.4466
p=0.2409
p=0.7461
p=0.8541
p=0.5555
p=0.8265
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2